

LAB SESSION 12

Programming Exercise:

Question 1: Write a class Circle with a private member radius. Write a friend function to calculate the area of the circle.

```
1  #include <iostream>
2  using namespace std;
3
4  class Circle
5  {
6  private:
7      double radius;
8
9  public:
10     // Constructor
11     Circle(double r)
12     {
13         radius = r;
14     }
15
16     // Friend function declaration
17     friend double calculateArea(const Circle &c);
18 };
19
20 // Friend function definition
21 double calculateArea(const Circle &c)
22 {
23     return 3.14159 * c.radius * c.radius;
24 }
25
26 int main()
27 {
28     Circle c1(5.0); // Circle with radius 5
29
30     // Call friend function to calculate area
31     cout << "Area of Circle: " << calculateArea(c1) << endl;
32
33     return 0;
34 }
```

```
PS C:\Users\hp\Unity Projects\Flappy Bird> cd "c:\C++ PROGRAMS\" ; if ($?) { g++ tut1.cpp -o tut1 } ; if ($?) { .\tut1 }
Area of Circle: 78.5397
PS C:\C++ PROGRAMS>
```

Question 2: Implement a class Rectangle with private members length and width. Write a friend function to calculate the perimeter of the rectangle.

```
C:\C++ PROGRAMS > C++ tut1.cpp > ...
3  class Rectangle
4  {
5  private:
6      double length;
7      double width;
8
9  public:
10     // Constructor
11     Rectangle(double l, double w)
12     {
13         length = l;
14         width = w;
15     }
16
17     // Friend function declaration
18     friend double calculatePerimeter(const Rectangle &r);
19 };
20 // Friend function definition
21 double calculatePerimeter(const Rectangle &r)
22 {
23     return 2 * (r.length + r.width);
24 }
25
26 int main()
27 {
28     Rectangle r1(5.0, 3.0); // Rectangle with length 5 and width 3
29
30     // Call friend function to calculate perimeter
31     cout << "Perimeter of Rectangle: " << calculatePerimeter(r1) << endl;
32
33     return 0;
34 }
35
```

PS C:\Users\hp\Unity Projects\Flappy Bird> cd "c:\C++ PROGRAMS\" ; if (\$?) { g++ tut1.cpp -o tut1 } ; if (\$?) { .\tut1 }
Perimeter of Rectangle: 16
PS C:\C++ PROGRAMS>